



Data Security

Secure Conference System

System Usage Manual

Goals:

The assignment aims at enforcing the encryption and communication techniques. It helps the student in acquiring the necessary knowledge in developing client/server application and in securing data transfer using cryptography techniques.



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Introduction:

Data security is the means of ensuring that data is kept safe from corruption and that access to it is suitably controlled. Thus data security helps to ensure privacy. It also helps in protecting personal data. Cryptography is the practice and study of hiding information. Modern cryptography intersects the disciplines of mathematics, computer science, and electrical engineering. Applications of cryptography include ATM cards, computer passwords, and electronic commerce.

Usage of the server application:

Click on Server.exe then this window below will appear to you. The main job for this window is:

- 1- Show the clients which are connected to the server.
- 2- Creates public and private key.
- 3- Receive Encrypted messages from clients. (Using DES algorithm)
- 4- Decrypt messages and forward it to all clients. (Using DES algorithm)



Configuration of the server application:

1. The administrator creates a public and private key for the server. The keys are stored on the server.

When the server.exe is run, server will create 2 files (public and private key) named **PublicKey.pke** and **PublicPrivateKeys.kez**.

2. The administrator creates also user accounts. The user information consists of the These data (username and the hash of the user password and the server public key) are stored in **UserConfiguration.xml** file.

3. The administrator launches the server and the server is waiting for connections from clients. The server listens to a certain IP address and a certain port.

The IP address and port are stored in **ServerConfiguration_IP_Port.xml** file.



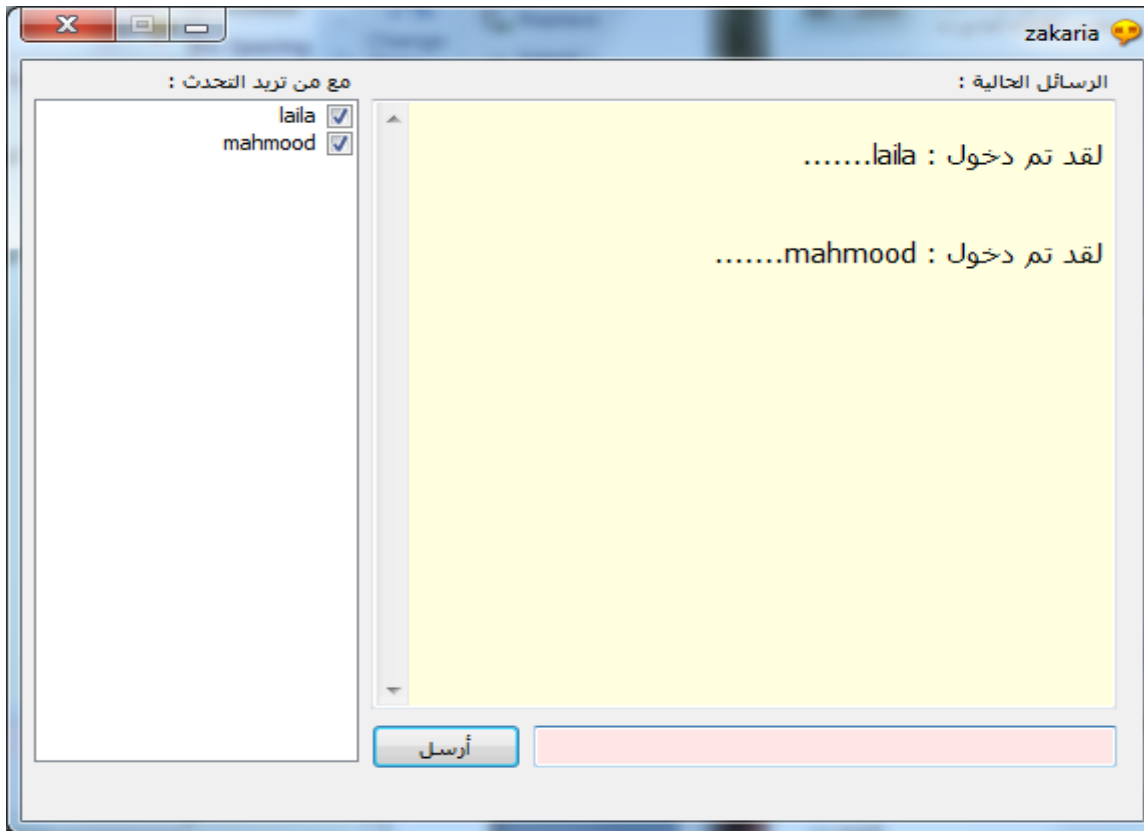
Usage of the client application:

First, Click on Client.exe then this window below will appear to the user. The main job for this window is:

- 1- The client shows IP address and Port from XML file which is configurable;
- 2- Write the user name and password then press on (تحقق) button.

A screenshot of a Windows-style application window titled "أدخل اسم المستخدم" (Enter username). The window has a light blue border and standard Windows window controls (minimize, maximize, close). Inside, there are two text input fields for "اسم المستخدم" (Username) and "كلمة المرور" (Password). Below these are two more input fields: "Port" with the value "1024" and "IP Address" with the value "127.0.0.1". At the bottom, there are three buttons: "إغلاق" (Close), "تسجيل دخول" (Login), and "تحقق" (Check/Verify). The "تحقق" button is highlighted with a blue border.

- 3- If the user name and password were correct, then (تسجيل الدخول) button will be enabled.
- 4- The user name is stored in XML file named Configuration, and hashing password is stored also in the same file.
- 5- The hashing password is encrypted **using MD5 Algorithm**.
- 6- When you write username and password, the client first search the username in the XML configuration file, if the username was not found, then error message will appear. If username was found then the client will encrypt the password using MD5 algorithm and compare the hashing password which stored in the file with the encrypted password. If it is same then (تسجيل الدخول) button will be enabled and you can log in to the client.
- 7- After that, press on (تسجيل الدخول) button to log in to this new window.



8- Then you can write to all users in the list.

Configuration of the client application:

- 1- When you start up the client application , you have to write username and password to compare it with the data stored in DB **UserConfiguration.xml** file.
- 2- When you join the meeting, you can select one user and talk to him, or you can select all users.
- 3- The client read the IP address and port from XML file and shows it on the client.
- 4- You can change IP and port from the file named: **ServerConfiguration_IP_Port.xml**

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